

Project Summary: Main Landing Gear - Bogie Beam Design & Structural Analysis

Boeing 777 · Personal Engineering Project

① Loads & Requirements

Loads derived from aircraft weight & landing g's; target factor of safety = 1.5.

Constants & Assumptions

Gravitational Constant	9.81
Maximum Landing Weight (MLW)	251,290.00
Main Gear Load Share %	90

Wheel Configuration

No. of Gear Legs	2.00
Wheels per Main Leg	6.00
Factor of Safety	1.50
Bogie-Beam Load Sharing	0.17

FORCE LOAD VALUES

Vertical Load Limit	2.50
Braking Coefficient of Friction	0.80
Side Load Factor (Crosswind)	0.50
Boeing Maximum Tyre Load	26,980.00

Factor of Safety will be performed in Fusion 360 Simulation Analysis
Assumes equal distribution of forces to all axle stubs of bogie-beam.
Maximum Value of Mass applied to tyres from BOEING safety standards

STATIC VALUES

Total Aircraft Weight	2,465,154.90
Vertical Load on main gear	1,232,577.45
Vertical Load per wheel	205,429.58

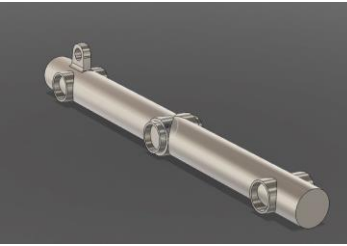
Main Gear Wheel Loads

Vertical F _z	513,573.94
Braking F _x	410,859.15
Crosswind F _y	102,714.79
Vertical F _z Ultimate Limit (FoS)	770,360.91
Braking F _x Ultimate Limit (FoS)	616,288.73
Crosswind F _y Ultimate Limit (FoS)	154,072.18

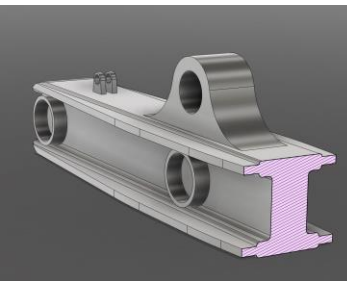
Total Gear Wheel Load

Vertical F _z	3,081,443.63
Braking F _x	2,465,154.90
Crosswind F _y	616,288.73
Vertical F _z Ultimate Limit (FoS)	4,622,165.44
Braking F _x Ultimate Limit (FoS)	3,697,732.35
Crosswind F _y Ultimate Limit (FoS)	924,433.09

② Beam Design Evolution



Cylinder: Poor placement of mass for reducing bending stresses

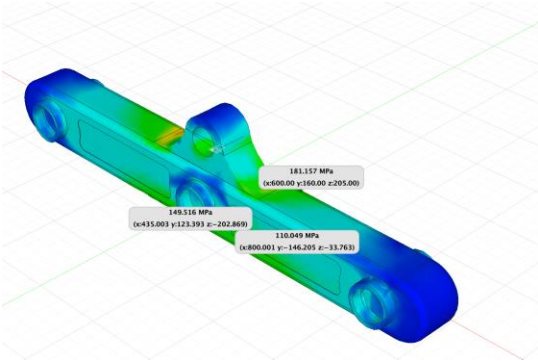


I-beam: Excellent support against vertical loading but weak when introduced to multi axial forces .

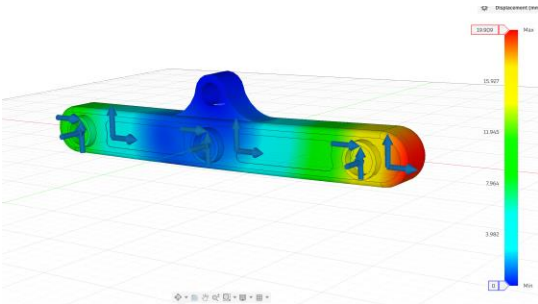


Final: hollow rectangular section. Significantly lower mass whilst retaining structural strength in all directions.

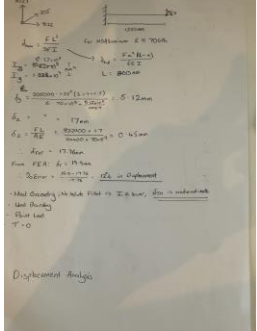
③ Structural Validation



Von Mises stress evaluated at 3 points randomly chosen along beam geometry.



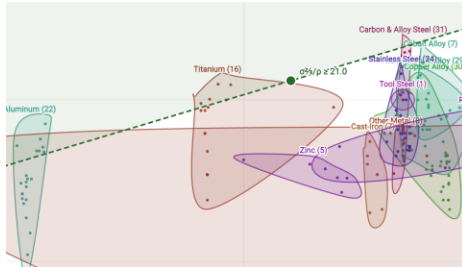
Displacement analysis (max 19.9 mm under ultimate load)



Example Hand Calculations.
Using Euler Bernoulli Beam Theory to predict displacement of the end of beam.

Other calculations were performed for Von Mises Stress

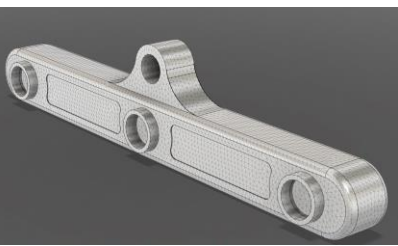
④ Materials & Mesh Convergence



Ashby Selection Criteria used to screen 3 possible materials. Choosing Ti-6Al-4V.



Lower Landing Gear CAD Assembly inc. Torque Link, OLEA Strut and Hydraulic Actuators.



Mesh Convergence study used to determine when FEA simulation is stable. Final mesh was deemed acceptable when results varied <2.5%.